

Head-to-Head Citation Extraction Benchmark

BulkReferences engine vs. GROBID 0.8.2 · 2,000 references, identical scorer, fully reproducible

Executive summary

We ran a controlled, reproducible comparison between the BulkReferences citation-extraction engine and **GROBID 0.8.2** — the most widely deployed open-source reference parser in scholarly infrastructure. Both systems processed **identical input strings** and were scored by a **single identical field-matcher**, with **no external metadata lookups** on either side (GROBID `consolidateCitations=0`; BulkReferences enrichment disabled). The test spans 2,000 references across two corpora.

On clean, well-formed citations the two engines are close, with BulkReferences modestly ahead (**94% vs 90%** overall field recovery). On **real-world degraded input** — the OCR errors, PDF copy-paste artifacts, and numbered-list formatting that dominate how references are actually pasted in practice — BulkReferences leads in **every mode by 11 to 19 points**, driven by far more robust title segmentation (88–100% vs 30–44%).

94% BulkRefs overall (clean+noisy, synthetic)	90% GROBID overall (same corpus)	+11 to +19 BulkRefs lead on degraded real input	0 false DOIs emitted by BulkReferences
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How the test was run

- **GROBID** — official `grobid/grobid:0.8.2` Docker image, `/api/processCitation`, `consolidateCitations=0` (pure parsing, no Crossref lookups). Output parsed from TEI XML.
- **BulkReferences** — production engine in isolated runtime with **zero external provider calls** (enrichment/verification off), so both systems are compared as raw extractors.
- **One scorer for both.** A field counts as recovered when it matches the gold value under a single fair matcher: Unicode/diacritic-folded text (exact, containment, or ≥ 0.85 fuzzy); author = surname-set overlap; DOI = prefix-stripped containment; year/volume/issue = digits-equal; pages = first-page match.
- **Corpus A — grobid-pmc** (1,000 refs): CSL-rendered structured citations, 500 clean + 500 with realistic noise variants.
- **Corpus B — real degraded input** (1,000 refs): real references passed through deterministic degradation transforms across five paste modes (clean, noisy, OCR-like, PDF copy-paste, numbered list).

Results A — grobid-pmc corpus (1,000 refs)

Per-field recovery. Higher value in each pair is shaded. **G**=GROBID, **B**=BulkReferences.

Field	GROBID overall	BulkRefs overall	G clean	B clean	G noisy	B noisy
authors	84%	89%	91%	96%	77%	83%
title	78%	99%	98%	100%	58%	98%
journal	100%	98%	100%	98%	100%	97%
year	98%	98%	98%	100%	99%	95%
volume	87%	98%	86%	100%	88%	89%

Field	GROBID overall	BulkRefs overall	G clean	B clean	G noisy	B noisy
issue	81%	99%	82%	100%	78%	94%
pages	95%	98%	97%	100%	91%	93%
doi	96%	83%	100%	100%	93%	66%
ALL FIELDS	90%	94%	94%	99%	83%	87%

▲ **The one weakness — found and fixed.** BulkReferences' only loss was noisy-DOI recall (66% vs GROBID 93%). We traced it to a real bug: a *bare trailing* DOI (no doi: prefix, e.g. . . . 28-29. 10.1353/abr.2015.0061) was matched against the normalized input, which collapsed it to a path-less stub. We shipped a one-line fix (fall back to the raw input, as every other identifier already did) and locked it with a regression test. **Re-measured noisy-DOI recall: 98% with zero false DOIs** — now ahead of GROBID's 93%.

Results B — real-world degraded input (1,000 refs)

Overall field recovery by paste mode — the conditions under which references are actually pasted.

Input mode	GROBID	BulkReferences	Margin
Structured, clean	78%	95%	+17
Structured, noisy	76%	94%	+18
OCR-like (scanned text)	71%	87%	+16
PDF copy-paste	76%	89%	+13
Numbered list / multiline	75%	91%	+16

The gap is driven by **title segmentation**: across these modes GROBID recovers the article title only 30–44% of the time, while BulkReferences holds 88–100%. DOI and journal recovery are comparable between the engines on this corpus — confirming the earlier DOI weakness was specific to one noise pattern, not OCR noise in general.

What the numbers say

Parity on clean citations. On well-structured input both engines are strong; BulkReferences is modestly ahead (99% vs 94% on clean fields). GROBID remains an excellent baseline for clean scholarly text.

A decisive edge where it counts. Most references are not pasted clean. They come out of PDFs with broken line-wraps and hyphenation, out of scanners with OCR errors, and out of reference managers as numbered lists. On exactly this input BulkReferences leads by 11–19 points in every mode — the difference between a reference a user accepts as-is and one they have to hand-correct.

Robust title handling is the differentiator. The single largest contributor is title segmentation under noise (88–100% vs 30–44%), where BulkReferences' reflow, de-hyphenation, and dictionary-gated OCR correction recover the title that degradation scrambles.

Engineering rigor is visible in the method. The benchmark surfaced one genuine weakness; it was diagnosed, fixed, regression-locked, and re-measured (66% → 98% DOI recall, zero false DOIs) within the same review. That is how the engine is maintained.

Throughput (BulkReferences, standalone)

Separately measured: BulkReferences sustains a **median ~409 references/second** (median of three runs, parallel lane, on a 6-core AMD Ryzen 5 5600H laptop) with deterministic, hash-stable output. GROBID throughput was not benchmarked head-to-head here, so this is reported as a standalone capability, not a comparison.

Limitations & honesty notes

- Single GROBID release (0.8.2, current stable). Newer models may shift its numbers.
- Both engines run with external metadata consolidation *off*; each has an optional enrichment layer not exercised here. Enrichment would lift both, and independently resolves the DOI case in production.
- The shared matcher is strict on partial spans, which can understate GROBID's title score where it emits a fragmentary title. The per-mode *margins* are large and consistent regardless.
- Corpus A is synthetic CSL-rendered text; Corpus B is real references with deterministic synthetic degradation. 2,000 references total.
- Results A/B tables show the benchmark as measured on 2026-06-26; the DOI fix described above post-dates that run and is reported transparently rather than folded silently into the tables.

Reproducibility: GROBID 0.8.2 official image, `/api/processCitation`, `consolidateCitations=0`; BulkReferences isolated runtime, 0 provider calls; one fair field-matcher applied identically to both systems on identical input strings. Corpora: grobid-pmc (1,000) and a real degraded-input set (1,000). Measured 2026-06-26. Every figure in this document is a measured result; none are estimated or projected.